Started on	Sunday, 12 February 2023, 8:34 PM
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Completed on	Sunday, 12 February 2023, 8:35 PM
Time taken	10 secs
Marks	0/30
Grade	**0** out of 100

Question 1

Not answered

Marked out of 1

Medical students have unmet needs in the areas of career and wellness advising. A study was conducted to describe the development of an Advisory College Program (ACP) and assess its effectiveness compared to a traditional one-on-one faculty advisor system. The ACP, consisting of four colleges co-led by Advisory College Directors and supported by key Faculty, was developed to provide structured career and wellness advising. In the ACP programme, students had several impromptu meetings with advisors and took part in college-wide activities aimed at wellness and career counselling. The authors compared the ACP to the former Faculty Advisory Program (FAP, in which each medical student is paired with a volunteer faculty member who serves as the student's advisor) using two parallel questionnaires. The FAP survey was provided to second- and third-year medical students at the beginning of the academic year 2007 - 2008 and consisted of 53 items regarding their experiences with the FAP the previous year before the ACP programme began. The ACP survey was delivered to all first-year students at the end of the academic year 2007 -2008 and was based on their experiences with the ACP. Surveys were completed by 74% of first-year students, 60% of second-year students and 88% of third-year students. Survey data demonstrated a significant increase in the number of students who could identify their advisor, the frequency of student-advisor contacts, and the perceived accessibility of advisors in the ACP compared to the FAP. While an ordinal logistic regression model did not demonstrate a significant effect of the new advising system on overall satisfaction, univariate analysis demonstrated a significant increase in student satisfaction with wellness and career counseling. Based on the analysis of the single question "I am satisfied with the overall advisory system,'' there was a statistically significant improvement in the ACP compared to the FAP (67% vs. 24%; $p<0.001$; SA - Strongly Agree; SD - Strongly Disagree). The ACP was more effective in promoting student wellness and career counseling than the traditional one-on-one faculty advisor system. Similar college-based programs may be beneficial to students at other medical school programs. (Sastre, E.A., et al. (2010). Medical Teacher, 32(10), e429-e435.). What is the purpose of this study?

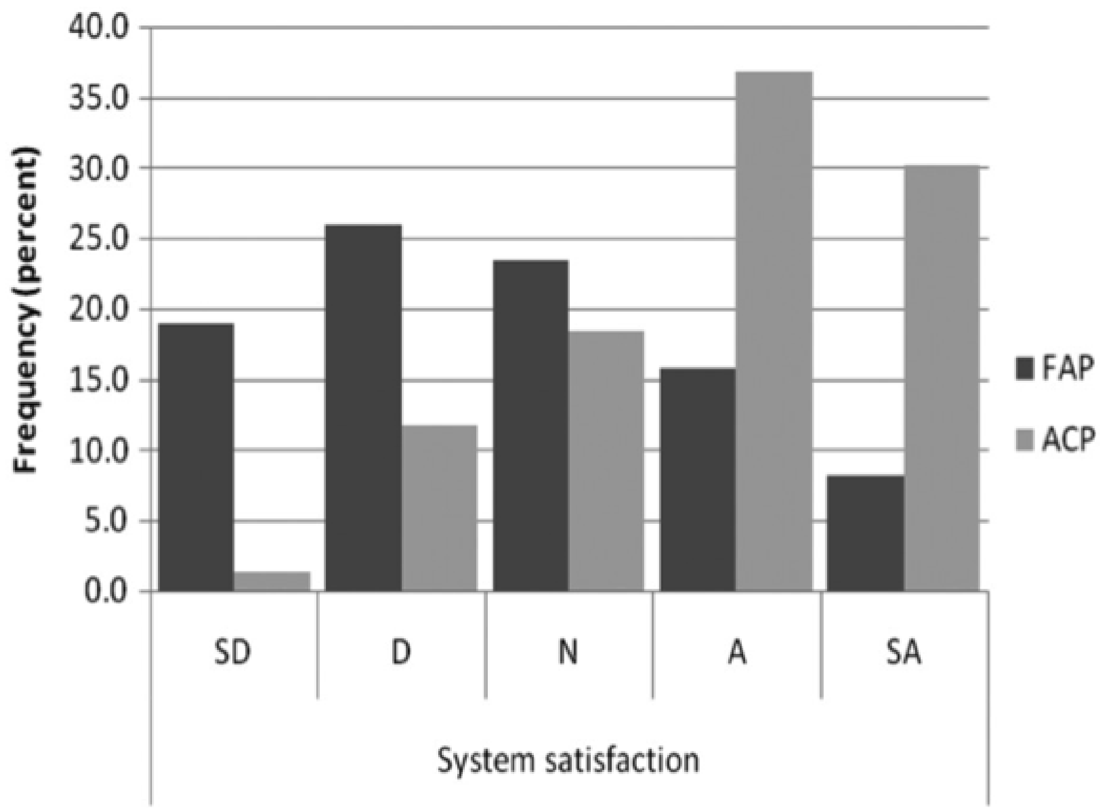

Figure 3. Distribution of students' responses to the question "I am satisfied with the overall advisory system" – "system satisfaction."

Select one:

- To measure the impact of a new advisory system (ACP) on the career success of a cohort of medical students.
- To compare the effectiveness of two methods of providing career and wellness advising to medical students
- To measure the improvement in satisfaction levels after introducing a new career advisory programme in a medical school
- To measure the unmet needs of medical students in the area of career advising

- To evaluate the student satisfaction with faculty advisor system

Your answer is incorrect.

While this study has been done in the context of unmet needs that medical students have in the areas of career and wellness advising, the primary objective of this study is to assess medical students' perceived effectiveness of the ACP system compared to the FAP system of career counselling.

The correct answer is: To compare the effectiveness of two methods of providing career and wellness advising to medical students

Question 2

Not answered

Marked out of 1

Medical students have unmet needs in the areas of career and wellness advising. A study was conducted to describe the development of an Advisory College Program (ACP) and assess its effectiveness compared to a traditional one-on-one faculty advisor system. The ACP, consisting of four colleges co-led by Advisory College Directors and supported by key Faculty, was developed to provide structured career and wellness advising. In the ACP programme, students had several impromptu meetings with advisors and took part in college-wide activities aimed at wellness and career counselling. The authors compared the ACP to the former Faculty Advisory Program (FAP, in which each medical student is paired with a volunteer faculty member who serves as the student's advisor) using two parallel questionnaires. The FAP survey was provided to second- and third-year medical students at the beginning of the academic year 2007 - 2008 and consisted of 53 items regarding their experiences with the FAP the previous year before the ACP programme began. The ACP survey was delivered to all first-year students at the end of the academic year 2007 -2008 and was based on their experiences with the ACP. Surveys were completed by 74% of first-year students, 60% of second-year students and 88% of third-year students. Survey data demonstrated a significant increase in the number of students who could identify their advisor, the frequency of student-advisor contacts, and the perceived accessibility of advisors in the ACP compared to the FAP. While an ordinal logistic regression model did not demonstrate a significant effect of the new advising system on overall satisfaction, univariate analysis demonstrated a significant increase in student satisfaction with wellness and career counseling. Based on the analysis of the single question "I am satisfied with the overall advisory system,'' there was a statistically significant improvement in the ACP compared to the FAP (67% vs. 24%; $p<0.001$; SA - Strongly Agree; SD - Strongly Disagree). The ACP was more effective in promoting student wellness and career counseling than the traditional one-on-one faculty advisor system. Similar college-based programs may be beneficial to students at other medical school programs. (Sastre, E.A., et al. (2010). Medical Teacher, 32(10), e429-e435.). What is the design employed in this study?

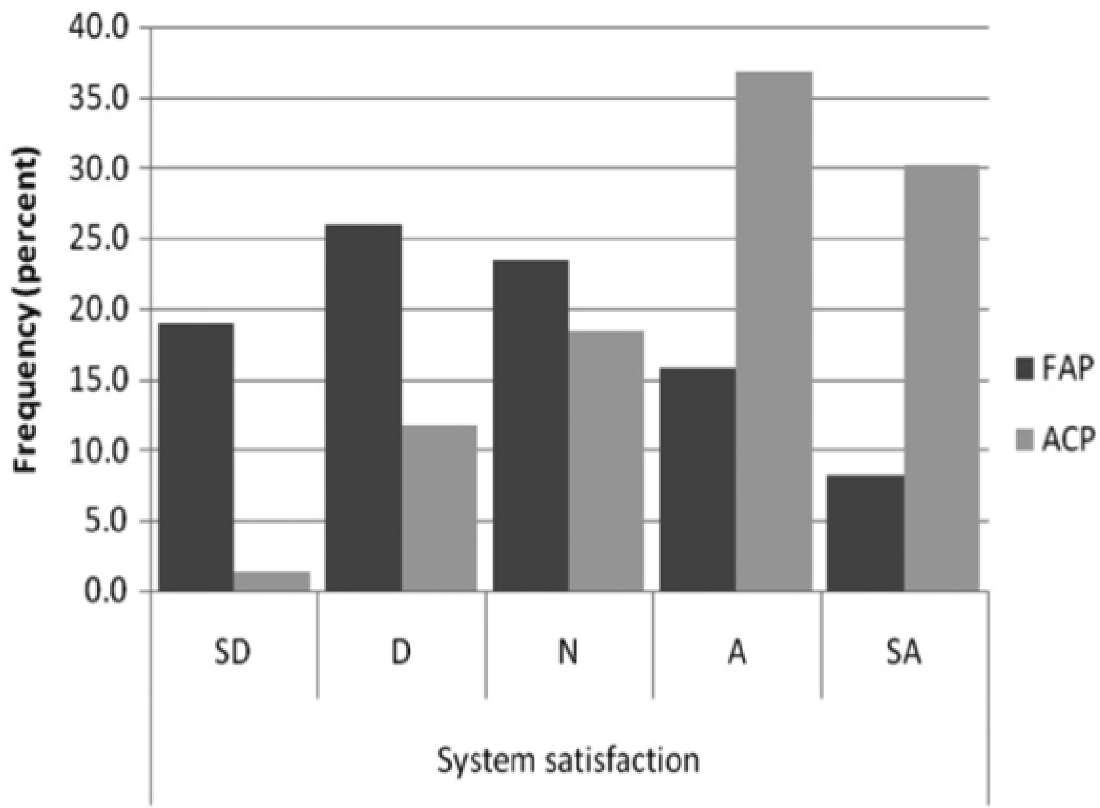

Figure 3. Distribution of students' responses to the question "I am satisfied with the overall advisory system" – "system satisfaction."

Select one:

- Historical controlled trial
- Cross sectional survey
- Qualitative research
- Before and after trial

- Uncontrolled trial

Your answer is incorrect.

As 2 different groups are compared on the outcome of an intervention, this study is best described as a controlled trial. In this particular instance, the controls are historical (those who had exposure to the compared intervention at sometime in the past, not in parallel with the evaluated intervention). A simple before-and-after study is one in which there is only one group receiving the intervention, but their historical data before the intervention serves as the control arm for comparison. An extension of a simple before and after design is the interrupted time series study. As the name suggests, this design uses measurements obtained from multiple time points in a group before and after an intervention (or 'interruption'). The objective is to test whether the intervention has had any effect on the rate of improvement (or worsening, e.g. in dementia) seen in the group. Another variation is the controlled before-and-after study where outcome measures are assessed before and after the implementation of an intervention (usually a policy change or service-level intervention) not in one, but TWO groups with one group receiving the new intervention and the other (control) group undergoing before-after measurements with either no intervening intervention or after a placebo. http://ccg.cochrane.org/non-randomised-controlled-study-nrs-designs

The correct answer is: Historical controlled trial

Question 3

Not answered

Marked out of 1

Medical students have unmet needs in the areas of career and wellness advising. A study was conducted to describe the development of an Advisory College Program (ACP) and assess its effectiveness compared to a traditional one-on-one faculty advisor system. The ACP, consisting of four colleges co-led by Advisory College Directors and supported by key Faculty, was developed to provide structured career and wellness advising. In the ACP programme, students had several impromptu meetings with advisors and took part in college-wide activities aimed at wellness and career counselling. The authors compared the ACP to the former Faculty Advisory Program (FAP, in which each medical student is paired with a volunteer faculty member who serves as the student's advisor) using two parallel questionnaires. The FAP survey was provided to second- and third-year medical students at the beginning of the academic year 2007 - 2008 and consisted of 53 items regarding their experiences with the FAP the previous year before the ACP programme began. The ACP survey was delivered to all first-year students at the end of the academic year 2007 -2008 and was based on their experiences with the ACP. Surveys were completed by 74% of first-year students, 60% of second-year students and 88% of third-year students. Survey data demonstrated a significant increase in the number of students who could identify their advisor, the frequency of student-advisor contacts, and the perceived accessibility of advisors in the ACP compared to the FAP. While an ordinal logistic regression model did not demonstrate a significant effect of the new advising system on overall satisfaction, univariate analysis demonstrated a significant increase in student satisfaction with wellness and career counseling. Based on the analysis of the single question ''I am satisfied with the overall advisory system,'' there was a statistically significant improvement in the ACP compared to the FAP (67% vs. 24%; $p<0.001$; SA - Strongly Agree; SD - Strongly Disagree). The ACP was more effective in promoting student wellness and career counseling than the traditional one-on-one faculty advisor system. Similar college-based programs may be beneficial to students at other medical school programs. (Sastre, E.A., et al. (2010). Medical Teacher, 32(10), e429-e435.). What is the power of the study?

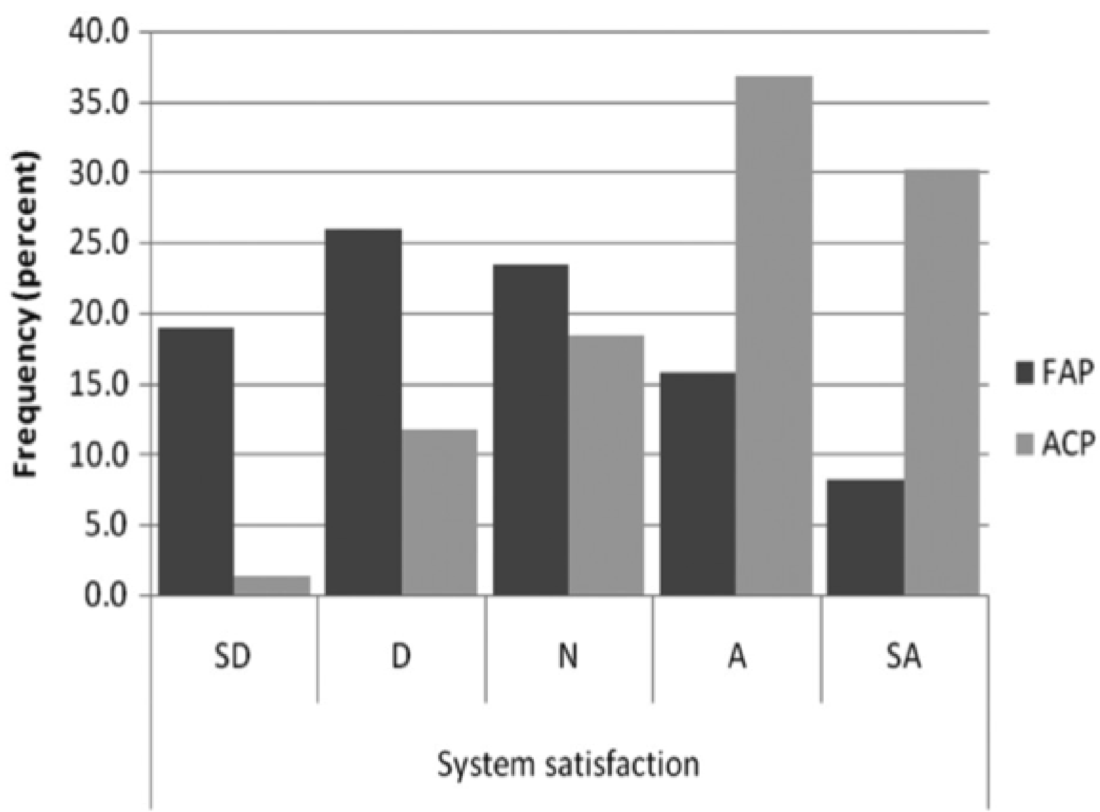

Figure 3. Distribution of students' responses to the question "I am satisfied with the overall advisory system" – "system satisfaction."

Select one:

- 60%
- 88%
- Cannot be determined from the given data
- 74%

- 74%

Your answer is incorrect.

The response rate for a survey is not same as the power. Power is the ability of a study to report a difference if that difference truly exists between two groups.

The correct answer is: Cannot be determined from the given data

Question 4

Not answered

Marked out of 1

Medical students have unmet needs in the areas of career and wellness advising. A study was conducted to describe the development of an Advisory College Program (ACP) and assess its effectiveness compared to a traditional one-on-one faculty advisor system. The ACP, consisting of four colleges co-led by Advisory College Directors and supported by key Faculty, was developed to provide structured career and wellness advising. In the ACP programme, students had several impromptu meetings with advisors and took part in college-wide activities aimed at wellness and career counselling. The authors compared the ACP to the former Faculty Advisory Program (FAP, in which each medical student is paired with a volunteer faculty member who serves as the student's advisor) using two parallel questionnaires. The FAP survey was provided to second- and third-year medical students at the beginning of the academic year 2007 - 2008 and consisted of 53 items regarding their experiences with the FAP the previous year before the ACP programme began. The ACP survey was delivered to all first-year students at the end of the academic year 2007 -2008 and was based on their experiences with the ACP. Surveys were completed by 74% of first-year students, 60% of second-year students and 88% of third-year students. Survey data demonstrated a significant increase in the number of students who could identify their advisor, the frequency of student-advisor contacts, and the perceived accessibility of advisors in the ACP compared to the FAP. While an ordinal logistic regression model did not demonstrate a significant effect of the new advising system on overall satisfaction, univariate analysis demonstrated a significant increase in student satisfaction with wellness and career counseling. Based on the analysis of the single question ''I am satisfied with the overall advisory system,'' there was a statistically significant improvement in the ACP compared to the FAP (67% vs. 24%; p<0.001; SA - Strongly Agree; SD - Strongly Disagree). The ACP was more effective in promoting student wellness and career counseling than the traditional one-on-one faculty advisor system. Similar college-based programs may be beneficial to students at other medical school programs. (Sastre, E.A., et al. (2010). Medical Teacher, 32(10), e429-e435.). With respect to the interventions used in this study, which of the following is an accurate statement?

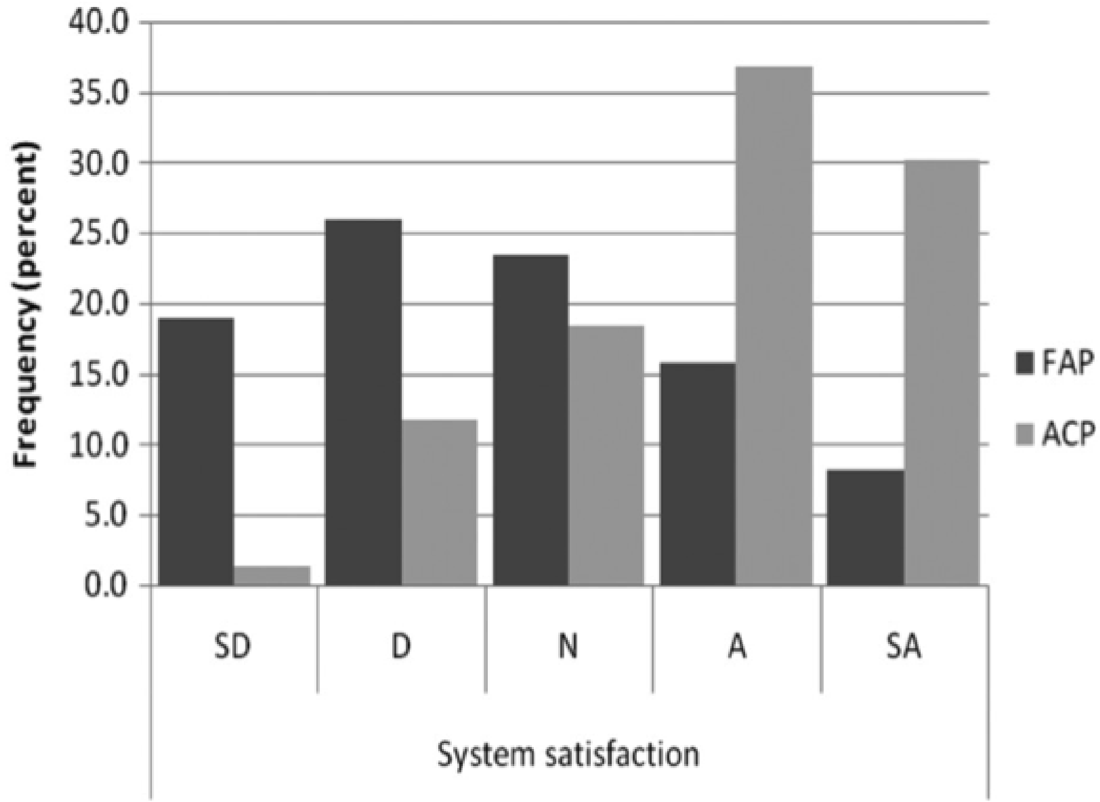

Figure 3. Distribution of students' responses to the question "I am satisfied with the overall advisory system" – "system satisfaction."

Select one:

- All participants were exposed to both FAP and ACP systems
- Only first year students were exposed to ACP system
- Second and third year students were exposed to both FAP and ACP systems
- Only second year students were exposed to ACP system

Only first year students were exposed to FAP system

Your answer is incorrect.

This study has used historical controls (i.e. a group who received the compared 'control' intervention FAP in the past (prior to 2007-08), before the new intervention ACP was introduced in 2007-08). Thus first year students were exposed only to ACP, while the second and third year students had exposure to both systems, though only their experience with FAP was evaluated in the current study. Instead of this designs, the authors could have chosen to compare the experience of ACP vs. FAP in a single set of students who had exposure to both systems. The latter approach (a form of before-after trial) could have potentially reduced the between-group variability thus increasing the internal validity.

The correct answer is: Second and third year students were exposed to both FAP and ACP systems

Question 5

Not answered

Marked out of 1

Medical students have unmet needs in the areas of career and wellness advising. A study was conducted to describe the development of an Advisory College Program (ACP) and assess its effectiveness compared to a traditional one-on-one faculty advisor system. The ACP, consisting of four colleges co-led by Advisory College Directors and supported by key Faculty, was developed to provide structured career and wellness advising. In the ACP programme, students had several impromptu meetings with advisors and took part in college-wide activities aimed at wellness and career counselling. The authors compared the ACP to the former Faculty Advisory Program (FAP, in which each medical student is paired with a volunteer faculty member who serves as the student's advisor) using two parallel questionnaires. The FAP survey was provided to second- and third-year medical students at the beginning of the academic year 2007 - 2008 and consisted of 53 items regarding their experiences with the FAP the previous year before the ACP programme began. The ACP survey was delivered to all first-year students at the end of the academic year 2007 -2008 and was based on their experiences with the ACP. Surveys were completed by 74% of first-year students, 60% of second-year students and 88% of third-year students. Survey data demonstrated a significant increase in the number of students who could identify their advisor, the frequency of student-advisor contacts, and the perceived accessibility of advisors in the ACP compared to the FAP. While an ordinal logistic regression model did not demonstrate a significant effect of the new advising system on overall satisfaction, univariate analysis demonstrated a significant increase in student satisfaction with wellness and career counseling. Based on the analysis of the single question ''I am satisfied with the overall advisory system,'' there was a statistically significant improvement in the ACP compared to the FAP (67% vs. 24%; p<0.001; SA - Strongly Agree; SD - Strongly Disagree). The ACP was more effective in promoting student wellness and career counseling than the traditional one-on-one faculty advisor system. Similar college-based programs may be beneficial to students at other medical school programs. (Sastre, E.A., et al. (2010). Medical Teacher, 32(10), e429-e435.). The proportion of students that strongly agree with the statement "I am satisfied with the overall advisory system" is

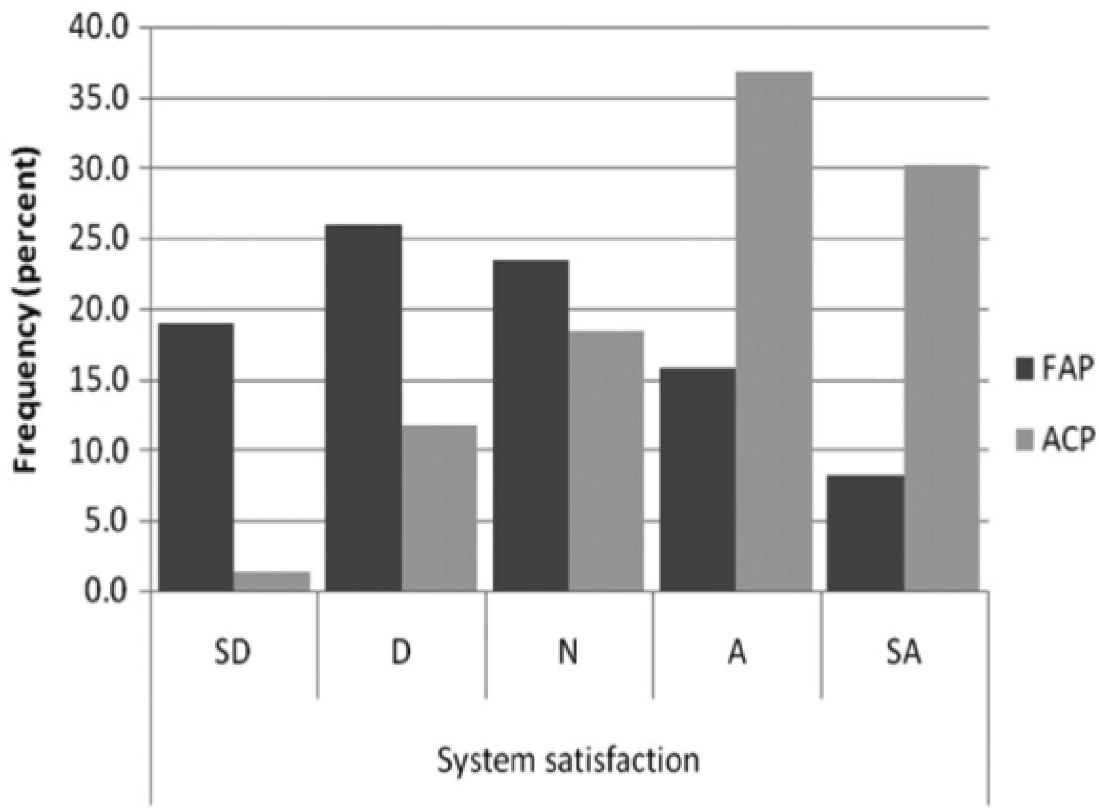

Figure 3. Distribution of students' responses to the question "I am satisfied with the overall advisory system" – "system satisfaction."

Select one:

- Significantly greater in the FAP group that ACP group
- Significantly greater in the ACP group that FAP group
- 30% in the ACP group
- 67% in the ACP group

24% in the FAP group

Your answer is incorrect.

The attached figure is a bar chart. The bar for 'strongly agree' option indicates 30% endorsement rate for the ACP group. Note that the result described in the text ("there was a statistically significant improvement in the ACP compared to the FAP 67% vs. 24%") does NOT refer to the proportion of those who endorse 'strongly agree'. Though not described in the given precis, it is likely that the authors collapsed the options 'agree' and 'strongly agree' in order to make a statistical comparison, giving the figures 67% and 24%.

The correct answer is: 30% in the ACP group

Question 6

Not answered

Marked out of 1

A researcher wants to make sure that the size of the control and two intervention groups are more or less equal so that statistical deductions can be meaningful. Which of the following methods of randomisation could ensure this?

Select one:

- Block randomisation
- Stratification
- Cluster randomisation
- Quasi-randomisation
- Simple randomisation

Your answer is incorrect.

Block randomization refers to a method where randomization occurs in a set of specified numbers, say blocks of 6 etc., to ensure equal number distribution between two arms. Larger the block size, lesser the ability to predict allocation. In cluster randomization, the unit of randomization and analysis is various centres or catchment-zones and not individual patients. Stratification is classifying to subgroups before randomising within each subgroup to ensure impartial allocation of subgroup factors. Quasi-randomisation refers to 'randomizing' using even/odd numbers of the date of birth, day of the week patient was seen, etc.

The correct answer is: Block randomisation

Question 7

Not answered

Marked out of 1

Which of the following is not a proper method of randomization?

Select one:

- Every 6 patients after recruitment are assigned according to a sequence; a new sequence is generated for the next 6.
- Everyone with odd number of birth date gets treatment, even number gets placebo; the allocation takes place even before examining a patient.
- A random list generated by computer is used to assign groups
- A random list generated using an excel sheet is administered by a third party
- 124 cities where a large RCT is planned are randomly allocated and every city allocated to one arm receives the same intervention/placebo

Your answer is incorrect.

Quasi-randomisation refers to randomizing using even/odd numbers of the date of birth, day of the week patient was seen, etc. These are not reproducible methods, and the sequences cannot ensure equal distribution of variables. These must be avoided. Computer generated lists are commonly used and are acceptable. 'Six patient sequences' refer to block randomization. Randomization of cities refers to cluster randomization.

The correct answer is: Everyone with odd number of birth date gets treatment, even number gets placebo; the allocation takes place even before examining a patient.

Question **8**

Not answered

Marked out of 1

To determine the efficacy of exposure therapy or cognitive restructuring relative to a wait-list control group in preventing chronic PTSD in those with acute stress reaction, a trial was designed. Participants were randomly allocated to 1 of 3 treatment conditions. Randomization was conducted by a process of stratification by sex, trauma type, and Acute Stress Disorder score. Random numbers system was administered by a third party in a distant centre not involved in the study. Which of the following is false with regard to the above study method?

Select one:

- All patients had equal chance of entering the waiting list group
- An attempt has been made to conceal the random allcoation
- Selection of patients who are less likely to develop PTSD to one group is avoided
- Probability theory can be used to make inferences.
- The baseline characters suspected to be important are randomly distributed

Your answer is incorrect.

The above design refers to RCT. So probability theory can be used (i.e. use of 'p-values'). Selection of a higher likelihood group to one side has been avoided by a process of stratification - so baseline characters are not truly random. This is preferred when the sample is small and will avoid skewed distribution of influential variables.

The correct answer is: The baseline characters suspected to be important are randomly distributed

Question 9

Not answered

Marked out of 1

To determine the efficacy of exposure therapy or cognitive restructuring relative to a wait-list control group in preventing chronic PTSD in those with acute stress reaction, a trial was designed. Participants were randomly allocated to 1 of 3 treatment conditions. Randomization was conducted by a process of stratification by sex, trauma type, and Acute Stress Disorder score. Random numbers system was administered by a third party in a distant centre not involved in the study. In the above study the reason for using a third party in assignment of random sequences is

Select one:

- To reduce the cost of the trial
- To ensure blinding
- To ensure allocation concealment
- To ensure stratification process
- To ensure subjects' compliance with intervention

Your answer is incorrect.

The reason for using a third party in the assignment of random sequences is to ensure allocation concealment. Allocation concealment refers to maintaining secrecy with regard to the allocation itself at the start of the study. Even if the study is randomized, at least one person in the trial team will have access to the generated random sequence and so can deduce where the next patient on the list will be allocated to. Concealing this information will avoid a selection bias wherein clinicians can pick up certain 'disease markers' that may make them favourably allocate a subject to one group or the other.

The correct answer is: To ensure allocation concealment

Question **10**

Not answered

Marked out of 1

Maintaining secrecy about the status of a patient as to which arm he or she belongs, once the allocation is complete is called

Select one:

- Allocation concealment
- Minimisation
- Stratification
- Blinding
- Restriction

Your answer is incorrect.

Blinding refers to maintaining secrecy about the status of a patient as to which arm he or she belongs, once the allocation is complete. If the participant only (or rarely therapist only) is not aware of what treatment he is receiving this is single blind. If both the participant and therapist are unaware, then this is double blind - this is the most common method. If the participant, therapist and the data analyzer - all three are unaware, this is called triple blind. If no one is blind, then it is called an open-label RCT.

The correct answer is: Blinding

Question 11

Not answered

Marked out of 1

Which of the following types of error is most obviously reduced by blinding in RCTs?

Select one:

- Referral bias
- Berkson bias
- Measurement bias
- Selection bias
- Confounding bias

Your answer is incorrect.

Blinding is necessary to obviate measurement bias - information bias, exposure suspicion/ascertainment bias (investigator having guessed who received the drug rather than placebo may unconsciously rate the improvement better).

The correct answer is: Measurement bias

Question **12**

Not answered

Marked out of 1

100 patients are enrolled in an RCT comparing an herbal medicine with fluoxetine at 12 weeks for depression. Out of 50 who were assigned to take fluoxetine, 8 did not turn up for follow-up. Out of 50 who were assigned to herbal treatment, 12 did not complete the treatment or missing. Which of the following is the best way of reporting the results of this RCT?

Select one:

- Report the number improved out of 42 for fluoxetine and 38 for herbal medicine
- Report the number improved out of 42 for fluoxetine and 50 for herbal medicine
- Report the number improved out of 50 for fluoxetine and 38 for herbal medicine
- This RCT should not be reported
- Report the number improved out of 50 for fluoxetine and 50 for herbal medicine

Your answer is incorrect.

The intention-to-treat principle holds that participants must be counted in the groups to which they were initially randomized, regardless of whether they were compliant with the allocated intervention. This includes participants who were withdrawn from active treatment for any reason. This is the best way of reporting a trial result in most instances. So reporting the number improved out of 50 for fluoxetine and 50 for herbal medicine is preferable.

The correct answer is: Report the number improved out of 50 for fluoxetine and 50 for herbal medicine

Question **13**

Not answered

Marked out of 1

100 patients are enrolled in an RCT comparing an herbal medicine with fluoxetine at 12 weeks for depression. Out of 50 who were assigned to take fluoxetine, 8 did not turn up for follow-up. Out of 50 who were assigned to herbal treatment, 12 did not complete the treatment or missing. If the above study reports numbers improved out of all 100 randomised, which of the following processes has been carried out?

Select one:

- Per protocol analysis
- Multiple regression analysis
- Stratification analysis
- Intention to treat analysis
- Treatment received analysis

Your answer is incorrect.

This is called intention to treat analysis. The rationale is that in most RCTs, one sets out to estimate the broader effects of allocating an intervention in clinical practice, rather than just the narrow effects in the subgroup of participants who adhere to it. The ITT strategy gives a conservative estimate of the treatment effect compared with what would be expected in an ideal world of full compliance. Hence, ITT tends to make RCT results more generalizable and pragmatic.

The correct answer is: Intention to treat analysis

Question **14**

Not answered

Marked out of 1

100 patients are enrolled in an RCT comparing an herbal medicine with fluoxetine at 12 weeks for depression. Out of 50 who were assigned to take fluoxetine, 8 did not turn up for follow-up. Out of 50 who were assigned to herbal treatment, 12 did not complete the treatment or missing. In the above study in order to calculate the possible missing values of 20 patients, the statistician uses the HAMD score recorded at the most recent interview for the lost patients and includes them as the final score in analysis. This approach is called

Select one:

- Worst case scenario analysis
- Best case scenario analysis
- Hot deck imputation technique
- Last observation carried forward
- Mean imputation technique

Your answer is incorrect.

This is called last observation carried forward LOCF method. There are many different ways by which one can substitute missing values when doing (ITT) intention to treat analysis. The LOCF is only useful for longitudinal observations where continuous scales were used. E.g. in a clinical trial of fluoxetine for depression observed for six months, the 3rd month's HAMD/MADRS values can be carried forward to 4th, 5th and 6th months and used in the final analysis. But many psychiatric diseases have a natural course of remission. LOCF does not make allowances for this. LOCF is 'too good' for controls and 'too stringent' for the patient group.

The correct answer is: Last observation carried forward

Question **15**

Not answered

Marked out of 1

100 patients are enrolled in an RCT comparing an herbal medicine with fluoxetine at 12 weeks for depression. Out of 50 who were assigned to take fluoxetine, 8 did not turn up for follow-up. Out of 50 who were assigned to herbal treatment 12 did not complete the treatment or missing. If such a trial is carried out to see the biological efficacy of the new herbal medicine and not to compare its clinical effectiveness against fluoxetine, which of the following methods of analysis will be most suitable?

Select one:

- Sensitivity analysis
- Mean imputation method
- Last observation carried forward analysis
- Intention to treat (ITT) analysis
- Per-protocol analysis

Your answer is incorrect.

ITT may not suit those who are investigating biological effects of a drug via RCT model - those who do not take the drug cannot have a biological effect. But for those studies that report on effectiveness, ITT is invaluable. Using per-protocol (PP) analysis only patients who sufficiently complied with the trial's protocol are considered in the final analysis. This is the antithesis of intention to treat analysis. It is also called 'oeon treatment'' analysis. PP analysis is often carried out in biological explanatory RCTs and as the secondary analysis in effectiveness trials.

The correct answer is: Per-protocol analysis

Question **16**

Not answered

Marked out of 1

Two educational interventions are compared with respect to efficacy in reducing psychosis related disability in high-risk individuals. The main outcome measured was median days of DUP (duration of untreated psychosis) reduced by each intervention. Which of the following is true concerning the above study method?

Select one:

- DUP is a secondary outcome measure
- DUP is a surrogate endpoint
- The study design is flawed
- This study cannot be a randomised controlled trial
- DUP is an irrelevant endpoint

Your answer is incorrect.

The researcher intends to demonstrate a reduction in psychosis related disability. But instead he/she chooses to measure median days of DUP, considering DUP as a proxy for disability related to psychosis. Hence, DUP is a surrogate endpoint. Secondary outcome measures are outcomes from testing an additional hypothesis different from the primary one.

The correct answer is: DUP is a surrogate endpoint

Question **17**

Not answered

Marked out of 1

In an allocation scheme, the next allocation of a subject depends on the characteristics of those already allocated. This scheme can be called

Select one:

- Matching scheme
- Blocking scheme
- Haphazard randomisation scheme
- Minimisation scheme
- Stratification scheme

Your answer is incorrect.

In minimisation schemes, next allocation depends on characteristics of those already allocated. Allocation of each participant aims to ensure a balance of prognostic factors between groups. The disadvantage is that this method is inferior to proper randomisation as it allocation is somewhat exposed and 'controlled' manually.

The correct answer is: Minimisation scheme

Question **18**

Not answered

Marked out of 1

A single subject is administered placebo and active intervention in tandem under double-blind controlled conditions. This type of study could be called

Select one:

- Before and after trial
- Cohort study
- N of 1 trial
- Cross over trial
- Uncontrolled trial

Your answer is incorrect.

N of 1 trial - here a single subject is administered placebo and active intervention (or two different interventions) in tandem under double-blind controlled conditions. This is the best way of optimizing a treatment schedule for an individual patient, but results cannot be generalized to other patients.

The correct answer is: N of 1 trial

Question **19**

Not answered

Marked out of 1

Which of the following factors seen in an RCT makes it different from natural clinical practice?

Select one:

- Strict exclusion criteria
- Being blind to treatment
- All of the listed options
- Frequent follow up and attention given to patients on trial
- Use of placebo

Your answer is incorrect.

Some of the practices while conducting RCTs that alienate then from normal clinical practice are as follows: 1. Recruitment: from specialist centres 2. Patients with co-morbid medical or psychiatric disorders are excluded 3. Patients are carefully selected to generate homogeneous diagnostic groups 4. Patients are allocated the treatment at random (no negotiation during prescription) 5. Patients are given detailed information for informed consent (rare in clinical practice) 6. Placebo is used to compare active treatment 7. Patients are followed at frequent intervals - often more frequent than clinical practice 8. Patients get detailed checklists of side-effects 9. Assessment end-point is typically 4-6 weeks 10. Often symptoms measures are the outcomes considered. 11. Patient and clinician are blind to treatment.

The correct answer is: All of the listed options

Question **20**

Not answered

Marked out of 1

Which of the following is the least adequate method of randomisation?

Select one:

- Simple randomisation by computer
- Toss of a fair, unbiased coin.
- Minimisation
- Odd/even last digit of date of birth
- Permuted block randomisation

Your answer is incorrect.

Quasi-randomisation refers to randomizing using even/odd numbers of the date of birth, day of the week patient was seen, etc. These are not reproducible methods, and the sequences cannot ensure equal distribution of variables. These must be avoided.

The correct answer is: Odd/even last digit of date of birth

Question **21**

Not answered

Marked out of 1

As regards pragmatic trials:

Select one:

- It is essential to have modelled all the details of the intervention beforehand
- The main aim is to tell us how well a drug works
- They have high external validity
- They have more exclusive referral criteria than explanatory trials.
- They are less generalisable than explanatory trials

Your answer is incorrect.

Pragmatic trials are more practice-related and naturalistic than experimental RCTs. Certain variables such as strict exclusion/inclusion criteria, dose of the drug to be used, time at which change or switch occurs, etc. are not controlled as tightly as in RCTs. The main aim is to see how effective an intervention will be in clinical practice, rather than demonstrating the efficacy of a biological compound. They have higher external validity and are more generalisable. Tunis, SR et al. Practical clinical trials. JAMA, 2003. 290, 12, 1624

The correct answer is: They have high external validity

Question **22**

Not answered

Marked out of 1

Concerning randomisation, choose the best option:

Select one:

- Non-randomised studies are less likely to be affected by confounding than randomised studies.
- It can only be performed at the level of individual patients
- It guarantees that the study will be free of confounders
- It automatically leads to poor generalisability
- Concealment of allocation of randomisation refers to whether the outcome of randomisation can be predicted

Your answer is incorrect.

Concealment of random allocation refers to whether the outcome of randomisation can be predicted or not. Randomisation can be done in clusters instead of individual patient units. Randomisation does not eliminate confounders fully; it only serves to eliminate their influence by equally distributing them across all groups. Randomisation is not always associated with poor generalisability; e.g. pragmatic RCTs are randomized but still have good generalisability. Altman DG and Schulz KF. Concealing treatment allocation in randomized trials. BMJ 2001;323:446-447

The correct answer is: Concealment of allocation of randomisation refers to whether the outcome of randomisation can be predicted

Question 23

Not answered

Marked out of 1

Which of the following is correct with respect to sample size in research studies?

Select one:

- We do not need to randomize if our sample size is sufficiently large
- A larger sample size is needed only for a larger population
- A large sample size ensures representativeness of the sample
- Irrespective of size, proper sampling techniques will ensure precision of estimates.
- Randomisation ensures precision of estimates.

Your answer is incorrect.

The necessity for randomisation does not depend on sample size, but in very small samples randomisation is difficult and can become meaningless. Irrespective of population size, a large sample size ensures representativeness to a certain degree. Precision is very much dependent on number of observations i.e. sample size. It is not ensured by mere randomisation.

The correct answer is: A large sample size ensures representativeness of the sample

Question **24**

Not answered

Marked out of 1

An RCT was terminated following an interim analysis as it was deemed unfair to continue placebo for one group of trial participants as the active arm showed huge beneficial effect. Which of the following is the likely problem with this practice of stopping trials earlier due to the observation of apparently large benefit?

Select one:

- Ethically unacceptable to stop RCTs midway
- Overestimation of treatment effect
- Underestimation of treatment effect
- Higher rates of attrition may occur
- Intention to treat analysis may not be possible

Your answer is incorrect.

Taking the point estimate of the treatment effect at face value may overestimate the actual effect if the trial is stopped early due to apparently high benefit. The smaller the number of events, the greater the risk that the play of chance will result in apparent effects far from the truth. If these studies are allowed to continue longer, the random high of treatment effects may even out. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008. Pg 154

The correct answer is: Overestimation of treatment effect

Question **25**

Not answered

Marked out of 1

In the traditional hierarchy of evidence in EBM for therapy, which of the following occupies the highest level

Select one:

- Cross sectional surveys
- Secondary research papers
- RCTs
- Case series
- Lab based research

Your answer is incorrect.

The generally agreed-upon hierarchies of evidence indicate that certain types of research results should be given more weight than other types. The specific hierarchy depends on the type of clinical question being asked. Results from a systematic review of RCTs should be given more weight than the results of a single RCT, and results from an RCT should be given more weight than results from uncontrolled or nonexperimental studies.

The correct answer is: Secondary research papers

Question **26**

Not answered

Marked out of 1

A published study reported the use of prospective data collected to compare the effectiveness and tolerability of divalproex and valproic acid in 9,260 psychiatric admissions. All participating clinicians were allowed to initiate treatment with either valproic acid or divalproex and switch medications without restrictions, on the basis of what they believed to be in the patients' best interest. This study is best described as

Select one:

- Quasi experimental study
- Controlled double blind cohort study
- Pragmatic RCT
- Case control study
- Cross over RCT

Your answer is incorrect.

A quasi-experiment is often used in the social sciences. ("Quasi" stands for likeness or resembling) Quasi-experiments share characteristics of true experiments that seek interventions or treatments. The key difference in this empirical approach is the lack of random assignment. In contrast, pragmatic RCTs are randomised.

The correct answer is: Quasi experimental study

Question **27**

Not answered

Marked out of 1

In an adequately randomized, well-concealed RCT evaluating the efficacy of depot risperidone against oral risperidone, the participants receiving oral risperidone were reviewed every four weeks in the clinic. The participants receiving depot were seen by community nurse every two weeks who administered the depot. The drop-out rates were comparable, and the outcomes were measured by involved clinicians using standardised tools. The result strongly favoured the use of the depot. Which one of the following bias is most likely to have occurred?

Select one:

- Detection bias
- Performance bias
- Selection bias
- Observer bias
- Exclusion bias

Your answer is incorrect.

Blinding is an attempt to prevent observer bias. If an interviewer or rater knows which treatment a patient is receiving, he or she may differentially inquire about certain symptoms or see improvement where none exists. Gray GE. Evidence-based psychiatry. American Psychiatric Publishing, 2004.Pg 58

The correct answer is: Observer bias

Question **28**

Not answered

Marked out of 1

Which of the following group involved in an RCT must be blind to treatment in order to avoid bias due to cointervention?

Select one:

- Clinicians
- Data collectors
- Adjudicators of outcome
- Patients
- Data analysts

Your answer is incorrect.

A clinician involved in a trial providing additional care (e.g., time, support, etc.) to patients in one treatment group but not to those in the other treatment group leads to cointervention or performance bias. In clinical trials, this effect can be minimized by blinding clinicians to the treatment being provided. Gray GE. Evidence-based psychiatry. American Psychiatric Publishing, 2004.Pg 57

The correct answer is: Clinicians

Question 29

Not answered

Marked out of 1

In a treatment trial that favoured active intervention against placebo, high number of subjects are lost to follow-up. Which of the following would reassure you that the validity of study results is not affected badly by the high attrition?

Select one:

- Best case scenario assumption does not change the inferences
- Less than 20% are lost to follow up
- The sample size is larger than 300
- There is comparable loss on both arms of the trial
- Worst case scenario assumption does not change the inferences

Your answer is incorrect.

One of the major sources of bias in cohort studies/RCTs concerns patients lost to follow-up. If patients lost to follow-up differ in outcomes from patients who remain in the study, estimates will be biased. This is called migration bias or attrition bias. One way of dealing with losses to follow-up in the analysis is to perform a best case/worst case analysis, a form of sensitivity analysis in which outcome statistics are first calculated assuming all of those lost to follow-up did well, then recalculated assuming all of those lost to follow-up had a bad outcome. A positive treatment trial that favours intervention, if withstands worst case assumption, then one can be reassured that the direction of results cannot be influenced by lost patients.

The correct answer is: Worst case scenario assumption does not change the inferences

Question **30**

Not answered

Marked out of 1

A doctor wants to test the efficacy of a combination of 2 antidepressants against an established antidepressant in treating depressed patients experiencing obsessive thoughts. She prescribes the combination for every alternate patient in his clinic while the other eligible patients are treated as usual. What kind of study design has she employed?

Select one:

- Pragmatic RCT
- Randomised controlled trial
- Naturalistic observational study
- Uncontrolled trial
- Nonrandomised controlled trial

Your answer is incorrect.

There are several ways in which an intervention can be evaluated. In addition to randomised controlled trials, experimental studies can be conducted with people being allocated to different interventions using procedures of sampling that are not random e.g. systematic consecutive sampling with every 2nd patient attending a clinic getting active drug while the others being given a placebo. Such trials can be termed non-randomized controlled trials. Note that these studies do have a control arm; but they are not randomly selected.

The correct answer is: Nonrandomised controlled trial